PeerLearn: Test Plan

PeerLearn Test Plan v1.1 (identifier TP-PL-01) explains what will be tested, how, by whom and when, and links every test to a requirement in the SRS. It follows the IEEE 829 test plan outline. Team: Amogh Shetty (PES1UG24AM034), Aarush Patil (PES1UG24AM005), Abheek Sen (PES1UG24AM009), Allen Anish (PES1UG24AM029).

Revision history. v1.0: version submitted for Part 1. v1.1 (31 Oct 2026): team members and test responsibilities named; test environment adds Docker, Docker Hub and Kubernetes; new test case TC-20 checks the deployed system.

1. Introduction

1.1 Purpose. To define the testing needed to show that PeerLearn meets the SRS and is secure enough to demo on 30 Oct 2026.

1.2 Scope. Testing covers the backend API, the database rules and the web frontend of PeerLearn. Email delivery is tested only up to the point where the system hands the message to the email service.

1.3 References. PeerLearn SRS v1.1 (requirement IDs FR, NFR, SR); PeerLearn Software Architecture and Design Specification v1.1 (component IDs CMP and API IDs); IEEE 829.

2. Test plan overview

Testing is done at three levels. **Unit tests** check one function, such as the match score calculation. **Integration tests** check that parts work together, such as the booking API with the database. **System tests** check a whole user scenario through the web interface or API. Unit and integration tests run automatically in GitHub Actions on every pull request.

3. Test items

The items under test are the parts of the software defined in the Architecture document.

Item	Components	Sprint delivered
Authentication and access control	CMP-02, CMP-03	1
Profiles, availability, requests and matching	CMP-04, CMP-05	1
Session booking and ratings	CMP-06	1 and 2
Study groups	CMP-07	2
Moderation, notifications and audit	CMP-08, CMP-09, CMP-10	2
Web frontend	CMP-01	1 and 2
Data layer and database	CMP-11, CMP-12	1

4. Features to be tested

Every requirement in the SRS is tested, except those marked below. The table groups requirements by feature area and names the test cases that cover them; the full test case table is in Section 9.

Feature area	Requirements	Test cases
Registration and login	FR-01, FR-02, SR-04	TC-01, TC-02, TC-03, TC-19
Role-based access	FR-02, SR-05	TC-04
Tutor subjects and availability	FR-03, FR-04	TC-05
Help request and matching	FR-05, FR-06, FR-07, FR-15	TC-06
Booking, notifications and privacy	FR-08, FR-11, FR-14, SR-03	TC-07, TC-08
Ratings	FR-09	TC-09
Study groups	FR-10	TC-10
Reports and moderation	FR-12	TC-11
Dashboard and responsive layout	FR-13, NFR-05	TC-17
Password and data protection	NFR-01, SR-01, SR-02	TC-12, TC-19
Performance	NFR-02, NFR-03	TC-18, TC-13
Abuse resistance	NFR-04, SR-08	TC-14
Input validation	SR-06	TC-15
Audit logging	NFR-06, SR-07	TC-16
Deployment (Docker, Docker Hub, Kubernetes)	SR-01, FR-02	TC-20

Features not tested by test cases: NFR-07 (availability) is checked from the hosting provider's uptime report after deployment, and NFR-08 (coverage) is checked from the coverage report in GitHub Actions.

5. Test approach

5.1 Security validation. Each security requirement in the SRS is checked by at least one test. These checks run mostly as automated API tests with Jest and Supertest, plus a short manual review of the code and database.

Check	How it is done	Requirement
Transport security	Send a request over plain HTTP and confirm it is rejected or redirected; confirm TLS 1.2 or higher.	SR-01
Password storage	Inspect the users table and confirm every hash is bcrypt with cost 10 or higher; confirm no API response	NFR-01, SR-02

Check	How it is done	Requirement
	contains a hash.	
Contact privacy	Request another user's profile before and after an accepted session and compare the fields returned.	SR-03, FR-14
Authentication and token expiry	Call a protected endpoint with no token, a wrong token and an expired token; expect HTTP 401.	SR-04
Role-based access	Call each Admin endpoint with a Learner token and a Tutor token; expect HTTP 403.	SR-05
Injection and input validation	Send SQL injection strings and over-long or wrong-type values in every input; expect a validation error and no data leak. Review code for parameterised queries.	SR-06
Audit log protection	Try to edit or delete log entries through the API; expect failure.	SR-07, NFR-06
Brute force and flooding	Make 6 failed logins and expect a 15-minute lock; send 101 requests in a minute and expect HTTP 429.	SR-08, NFR-04
Vulnerable libraries	Run npm audit in GitHub Actions; no critical or high issues may remain open.	All

5.2 Functional testing. Black-box tests of each functional requirement using normal input, boundary values (for example a 30-minute and a 121-minute availability slot) and invalid input.

5.3 Non-functional testing. Performance tests use a script that seeds 1,000 users and 200 Tutors and measures response times (NFR-02, NFR-03). Usability and responsive layout are checked manually at 360 pixel and desktop widths with at least 3 teammates (NFR-05).

5.4 Pass and fail criteria. A test case passes when the actual result equals the expected result. The release passes when all Must test cases pass, at least 90% of Should test cases pass, and no critical or high defect is open.

5.5 Suspension and resumption. Testing pauses when the build fails or the test database is unavailable, and resumes once the build is green or the database is restored.

6. Test environment and tools

Node.js backend, PostgreSQL test database seeded by script, Jest and Supertest for automated tests, a browser with developer tools for frontend checks, and GitHub Actions to run tests on every pull request.

Docker and Docker Compose run the full stack locally; Docker Hub stores the images built by CI; a kind Kubernetes cluster in GitHub Actions is used for the deployment test TC-20.

7. Responsibilities and schedule

Who	Test responsibility	When
Aarush Patil	Authentication, access control and security tests	Sprint 1 and 2
Amogh Shetty	Matching, group and performance tests	Sprint 1 and 2
Abheek Sen	Frontend, responsive and usability checks	Sprint 1 and 2
Allen Anish (test lead)	Automation in GitHub Actions, defect log, final test report	19 to 30 Oct

Test cases are written before 12 Oct; unit tests are built with the code in Sprint 1 (19 to 23 Oct); security and integration tests are finished in Sprint 2 (26 to 30 Oct).

8. Risks

Risk	Effect	Plan
Only one week per sprint	Some tests skipped	Automate the Must test cases first
Test database differs from production	Missed defects	Use the same PostgreSQL version and seed script
Late changes to requirements	Tests out of date	Update the traceability tables whenever an SRS ID changes

9. Test cases

There are **20 test cases** covering all 15 functional requirements, NFR-01 to NFR-06, all 8 security requirements and the deployed system, which is above the minimum of 7 to 10. Update this table as defects and changes appear.

ID	Covers	Type	Scenario	Steps	Expected result
TC-01	FR-01	Functional	Valid registration	Send name, ana@pes.edu and a 10-character password to POST /auth/register	201 Created; the new user can log in; the response has no password
TC-02	FR-01	Functional	Non-university email	Register with ana@gmail.com	400 VALIDATION_ER

ID	Covers	Type	Scenario	Steps	Expected result
					ROR; no account is created
TC-03	FR-02, SR-04	Security	Login success and failure	Log in with the correct password, then with a wrong one	Correct: 200 and a token. Wrong: 401 and no token
TC-04	FR-02, SR-05	Security	Role restriction	Call PATCH /admin/reports/{id} with a Learner token	403 FORBIDDEN; the report is unchanged
TC-05	FR-03, FR-04	Functional	Subjects and availability	As a Tutor add Mon 16:00 to 17:00, then Mon 16:30 to 17:30, then an 11th subject	First slot 201; overlapping slot 409 SLOT_OVERLAP; 11th subject rejected with an error
TC-06	FR-05, FR-06, FR-07, FR-15	Functional	Ranked matches	Seed 7 Tutors for Maths (one has no Maths, one has 3 ratings); create a Maths help request; call GET /requests/{id}/matches	Exactly 5 Tutors, in descending score, none without Maths, no score above 100, rating bonus applied only to the rated Tutor
TC-07	FR-08, FR-11, FR-14, SR-03	Functional	Booking and contact privacy	Learner requests a slot; Learner views the Tutor profile; Tutor accepts; Learner views the profile again	Before acceptance email and phone are absent; after acceptance they are present; both users receive a notification
TC-08	FR-08, FR-11	Functional	Request expiry	Create a pending request, move the test clock	Status becomes expired and the Learner is notified

ID	Covers	Type	Scenario	Steps	Expected result
				forward 49 hours, run the expiry job	
TC-09	FR-09	Functional	Rating rules	Rate a pending session; rate a completed session with 5; rate it again; rate another with 6	Pending: rejected. Score 5: 201. Second rating: 409. Score 6: 400
TC-10	FR-10	Functional	Group capacity	Create a group of size 3; 4 Learners try to join; one member leaves	First 3 join; the 4th gets 409 GROUP_FULL; after a member leaves the 4th can join
TC-11	FR-12	Functional	Report and suspend	Learner reports a Tutor; Admin suspends the Tutor	Report is stored; the suspended Tutor cannot log in; the Admin action is logged
TC-12	NFR-01, SR-02	Security	Password storage	Register a user, read the users table, call every profile endpoint	password_hash starts with $2b$ and cost is 10 or more; no response contains a hash
TC-13	NFR-03	Performance	Matching speed	Seed 1,000 users and 200 Tutors; run 20 match requests	Every response arrives in under 2 seconds
TC-14	NFR-04, SR-08	Security	Rate limit and lockout	Enter a wrong password 5 times, then the correct one; send 101 requests in one minute with a valid token	The account is locked (423) for 15 minutes; the 101st request gets 429
TC-15	SR-06	Security	Injection and validation	Put ' OR '1'='1 in the email and topic fields;	400 VALIDATION_ERROR or no

ID	Covers	Type	Scenario	Steps	Expected result
				send a 10,000-character topic	match; no data is leaked; tables are unchanged
TC-16	NFR-06, SR-07	Security	Audit log	Log in, book a session, perform an Admin action; try to edit and delete log rows through the API and with the application database user	3 records with user, time and action; edit and delete are refused
TC-17	FR-13, NFR-05	Usability	Dashboard and responsive layout	Open the dashboard at 360 and 1280 pixels wide; count clicks to book a session	Upcoming sessions and pending requests (and a Tutor's rating) are shown; no horizontal scroll; at most 3 clicks
TC-18	NFR-02	Performance	API response time	Run 100 concurrent users on GET /dashboard and POST /requests for 1 minute	At least 95% of responses take under 1 second
TC-19	SR-01, SR-04	Security	HTTPS and token expiry	Request an http:// URL; call the API with a 3-hour-old token; call it with no token	Plain HTTP is rejected or redirected and TLS is 1.2 or higher; the old token gets 401 TOKEN_EXPIRED; no token gets 401
TC-20	SR-01, FR-02	System (deployment)	Deployed stack on Kubernetes	On a push to main, CI builds both images, pushes them to Docker Hub, deploys them to a kind	All pods ready and the migration Job complete; the home page returns PeerLearn; login

ID	Covers	Type	Scenario	Steps	Expected result
				Kubernetes cluster with k8s/deploy.sh, then opens the home page and logs in as Admin through nginx over HTTPS	returns a token

Coverage check: FR-01 to FR-15, NFR-01 to NFR-06 and SR-01 to SR-08 each appear in at least one test case. NFR-07 and NFR-08 are checked from reports, as stated in Section 4. TC-20 checks the deployed system end to end.